



Indraprastha Institute of Information Technology Delhi

(IIIT-Delhi), New Delhi

Department of Computer Science and Engineering

Course: Applied Cryptography (CSE556)

Project-2 Report

Learning Reduced-Round Cipher Behavior using Machine Learning

Submitted by:

Saloni MT25081

Amisha Patel MT25061

1. Abstract

Cryptographic algorithms are designed to survive different attacks, including those using machine learning and statistics-based attacks. Cryptographic cyphers with fewer rounds, however, may leak trends that can be learned by machine learning techniques. In this research, machine learning techniques were used to investigate the learnability of reduced round cryptographic cyphers.

Some of the eight cyphers evaluated included AES, SIMON, TinyJAMBU, Trivium, XOODOO, PRESENT, KATAN, and SPECK. Five machine learning models, including Logistic Regression, Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), Mutual Information Neural Estimator (MINE), and Random Forest, were used in the research.

Reduced rounds, including Rounds 1 to 5, were used to generate random plaintext data. From the plaintext data, the machine learning models were trained to predict the ciphertext. This shows that the accuracy of the predictions approaches random guessing as the number of rounds increases. It is evident that the more the number of rounds in the cypher, the greater the resistance to machine learning attacks and the spread.

2. Introduction

Sensitive data security and secrecy are ensured through modern cryptography techniques. To prevent intruders from identifying significant patterns, these algorithms utilize properties such as confusion and dispersion. However, currently, it is possible to scrutinize cryptography systems and uncover underlying statistical links, thanks to the discovery of machine learning algorithms.

A new field of research known as "machine learning-based cryptanalysis" focuses on investigating if patterns can be identified within encrypted data through learning algorithms. Reduced-round cryptography cyphers have the potential to reveal identifiable patterns since they lack dispersion. Reduced-round cyphers enable scrutiny of the efficiency of dispersion and cypher strength.

Cipher used:

- AES
- SIMON
- TinyJAMBU
- Trivium
- XOODOO
- PRESENT

- KATAN
- SPECK

Five machine learning models were used for analysis:

- Logistic Regression
- Multi Layer Perceptron (MLP)
- Convolutional Neural Network (CNN)
- MINE (Mutual Information Neural Estimator)
- Random Forest

3. Problem Statement

The purpose of using cryptographic cyphers is to produce random and unpredictable outputs. Reduced rounds of cypher implementations, however, could not provide sufficient dispersion and confusion, which might lead to statistical trends in the ciphertext.

The main objective :

- Can reduced-round cypher behaviour be learned by machine learning models??
- How does cypher learnability alter when the number of rounds increases??
- Which machine learning model is most effective for learning cryptography??
- Which cypher is most resistant to assaults based on machine learning??

Several machine learning models were trained on reduced-round ciphertext data in order to address these problems, and accuracy and Hamming distance metrics were used to assess the models' performance.

4. Methodology

Methodology used:

1. Datasets of plaintext were created at random.
2. Encryption using a reduced-round cypher was used.
3. Pairs of plaintext and ciphertext were produced.
4. Training and testing sets were created from the dataset.
5. Five machine learning models were trained:
 - Logistic Regression
 - MLP

- CNN
 - MINE
 - Random Forest
6. Models predicted ciphertext bits from plaintext inputs.
 7. Performance evaluated using:
 - Bitwise Accuracy
 - Hamming Distance
 8. Results compared across models and cipher rounds.

5. Machine Learning Models

we have used 5 machine learning.

- Logistic Regression
- Multi-Layer Perceptron (MLP)
- Convolutional Neural Network (CNN)
- MINE (Mutual Information Neural Estimator)
- Random Forest

5.1 Logistic Regression

Logistic regression is the supervised machine learning algorithm which estimates probabilities using sigmoid function. It finds if the linear pattern exists in cipher output.

5.2 Multi-Layer Perceptron (MLP)

MLP is feed forward neural network consisting of multiple layers. Each layer learns a complex layer. Encryption transformations are non linear, MLP are good in analysis as it learn precisely.

5.3 Convolutional Neural Network (CNN)

A CNN is deep learning design to process grid images. It checks for repeated patterns in cipher output bits by scanning through the output, applying some filter.

5.4 MINE (Mutual Information Neural Estimator)

MINE calculates how much similarity exists between plain text and cipher text, instead of classifying data. It generates connections between plain text and cipher text. If connection is 0 then no data can be learned.

5.5 Random Forest

The random forest is a supervised machine learning model that assembles multiple decision trees to generate an accurate output. It is good for complex analysis.

5.6 Evaluation Metrics

Two primary metrics were used to evaluate model performance:

- **Bitwise Accuracy (%)**: Fraction of correctly predicted ciphertext bits. An accuracy of 50% corresponds to random guessing; higher values indicate the presence of learnable patterns.
- **Hamming Distance (HD)**: Average number of bit differences between the predicted and actual ciphertext. Lower HD indicates better prediction quality. As HD approaches half the block length, cipher resistance improves.

5.7 Methodology

For each cipher, the following procedure was followed:

1. Generated a random plaintext dataset.
2. Applied reduced-round cipher encryption (Rounds 1 through 5).
3. Split the resulting data into training and testing sets.
4. Trained each of the five machine learning models.
5. Measured bitwise accuracy and Hamming distance on the test set.
6. Compared performance across models and rounds.

6. AES Cipher Analysis

6.1 About AES

Advanced Encryption Standard (AES) is a symmetric key block cipher that is widely used for secure data encryption. AES uses a 128-bit block and has a number of rounds of substitution, permutation, and mixing of keys. AES is a substitution permutation network-based cipher and has strong diffusion and confusion properties. Because of its complex design and the use of many non-linear transformations, AES is considered a secure cipher against conventional attacks. In the above project work, AES has been chosen as a strong cipher for comparison with lightweight crypto techniques.

6.2 Results

Table 1: AES – Bitwise Accuracy (%) across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	70.25	63.35	57.81	53.57	50.45
MLP	70.30	63.31	57.85	53.58	50.33
CNN	51.25	50.86	50.24	50.32	49.92
MINE	63.47	60.32	56.79	53.48	50.38
Random Forest	67.26	62.60	57.68	53.37	50.22

Table 2: AES – Hamming Distance across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	38.08	46.92	54.01	59.43	63.42
MLP	38.02	46.96	53.95	59.42	63.57
CNN	62.41	62.90	63.70	63.59	64.10
MINE	46.76	50.79	55.31	59.55	63.51
Random Forest	41.91	47.87	54.17	59.69	63.72

6.3 Plots

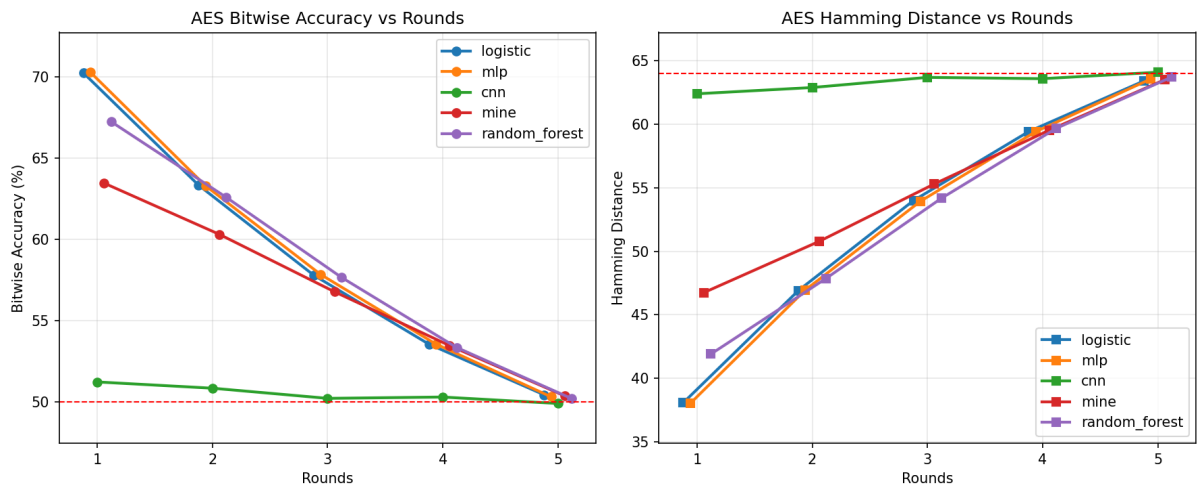


Figure 1: AES Accuracy vs Rounds and Hamming Distance vs Rounds

6.4 Observation

From the results, it can be concluded that the more the rounds in the AES algorithm, the lower the accuracy. In addition, the results show that the accuracy of all the models reduces to below 50%, i.e., the accuracy becomes random guessing. In addition, it can be noticed from the results obtained for the fifth round that all the models are performing at

a level close to 50%. This means that the AES algorithm becomes resistant to machine learning-based attacks. The Hamming distance also shows an increase corresponding to the rounds. This means that the predictions are becoming less accurate. The CNN model is performing at a level close to the random guessing level for all the rounds. This means that it becomes difficult to learn the structure of the AES algorithm. The MLP model is performing slightly better, but this comes at the expense of the training time.

6.5 Analysis

As the rounds increase, AES becomes harder to learn. There is some learnability in reduced-round AES. The AES was expected to be very secure. MLP is the worst in terms of speed, but it performs the best. To our surprise, logistic regression works well. Random forest performs moderately

6.6 Conclusion

AES has strong resistance to machine learning-based cryptanalysis. When model performance becomes closer to random guessing as rounds increase, AES security features are validated.

7. SIMON Cypher Analysis

7.1 About SIMON

SIMON is a lightweight block cipher that was created by the National Security Agency (NSA) of the USA. The SIMON block cipher is a block cipher that is used in embedded systems and IoT devices. The SIMON block cipher uses basic operations such as XOR, AND, and rotate operations. Unlike the AES encryption standard, which uses substitution-permutation encryption technique, the SIMON block cipher uses a Feistel structure. Therefore, it is computationally efficient but also good to machine learning attacks. In addition, the SIMON block cipher uses a number of operations to strengthen security.

7.2 Results

Table 3: SIMON – Bitwise Accuracy (%) across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	85.25	60.88	56.26	53.16	50.16
MLP	93.65	85.02	60.81	53.26	50.04
CNN	62.20	54.23	53.15	52.11	51.47
MINE	84.29	59.44	56.04	54.64	53.09
Random Forest	88.93	70.24	59.02	54.80	53.13

Table 4: SIMON – Hamming Distance across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	4.72	12.52	14.00	14.99	15.95
MLP	2.03	4.80	12.54	14.96	15.99
CNN	12.10	14.65	14.99	15.33	15.53
MINE	5.03	12.98	14.07	14.51	15.01
Random Forest	3.54	9.52	13.11	14.46	15.00

7.3 Plots

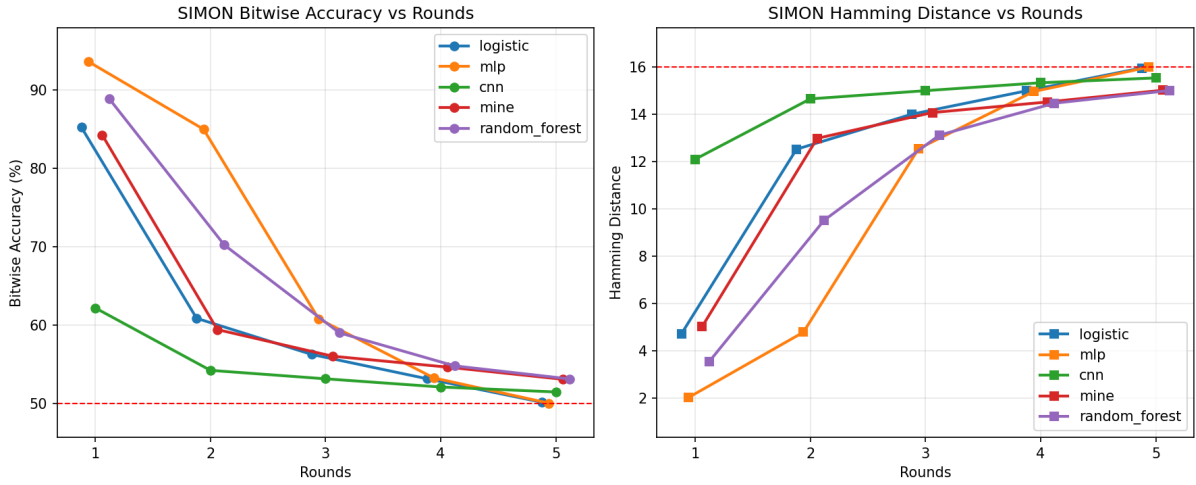


Figure 2: SIMON Accuracy vs Rounds and Hamming Distance vs Rounds

7.4 Observation

In the case of SIMON, learnability is very high in the early rounds. In the first round, the MLP model achieves 93.65% accuracy, which is much higher than AES. Other models achieve high accuracy as well. As the rounds increase, the accuracy of the models starts falling rapidly. In the fifth round, the accuracy is just 50%, which means the model is just guessing. This proves that the diffusion rate in SIMON is very low, which is why the models learn very easily. The Hamming distance is also increasing as the rounds increase, which proves that it is getting harder and harder to predict.

7.5 Analysis

SIMON highly learnable in reduced rounds. With rounds, accuracy progressively decreases. ML can learn a lightweight structure more easily. MLP best performing model. Logistic regression is quick and efficient. CNN performs moderately. Random forest performs consistently

7.6 Conclusion

The SIMON cypher is easier to learn than AES in less cycles. However, as the number of rounds increases, SIMON becomes more secure. This tells how round count significantly affects cypher security.

8. TinyJAMBU Cipher Analysis

8.1 About TinyJAMBU

TinyJAMBU is a lightweight encryption scheme. This scheme is for constrained environments such as IoT devices and embedded devices. TinyJAMBU is a finalist for the Lightweight Cryptography competition held by NIST. TinyJAMBU is based on a nonlinear feedback shift register scheme. This algorithm is based on bitwise operations such as XOR, AND, and shift operations. This algorithm is lightweight and may produce patterns that a machine learning algorithm could learn. TinyJAMBU is based on a number of rounds. The more the number of rounds, the better it is.

8.2 Results

Table 5: TinyJAMBU – Bitwise Accuracy (%) across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	71.87	64.10	57.86	52.83	49.80
MLP	71.54	64.06	58.01	53.14	49.99
CNN	56.05	53.56	52.14	50.83	50.07
MINE	66.39	61.74	56.86	52.83	50.27
Random Forest	71.89	64.10	57.88	53.22	50.09

Table 6: TinyJAMBU – Hamming Distance across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	18.00	22.98	26.97	30.19	32.13
MLP	18.21	23.00	26.87	29.99	32.01
CNN	28.13	29.72	30.63	31.47	31.95
MINE	21.51	24.49	27.61	30.19	31.83
Random Forest	17.99	22.98	26.96	29.94	31.94

8.3 Plots

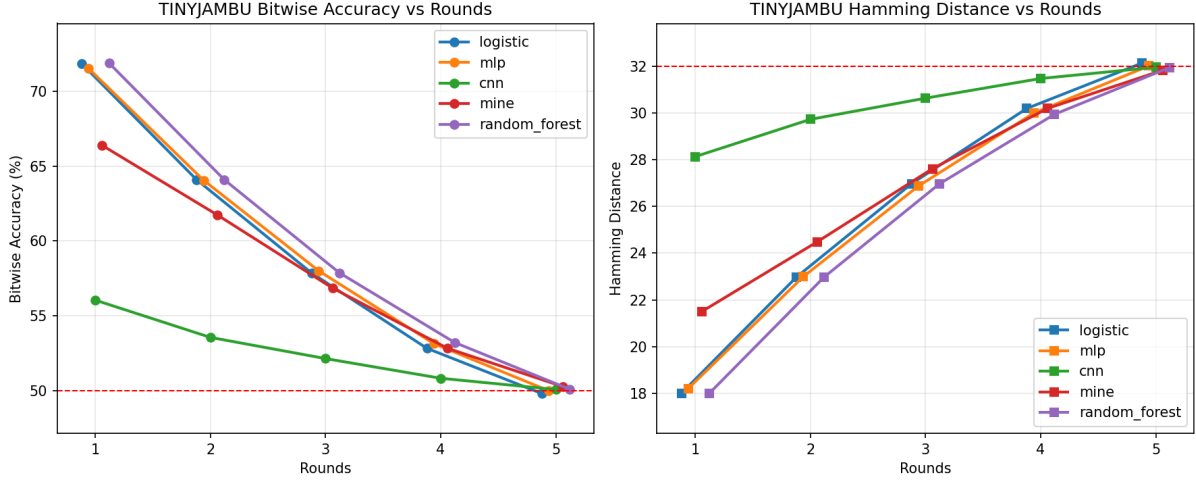


Figure 3: TinyJAMBU Accuracy vs Rounds and Hamming Distance vs Rounds

8.4 Observation

TinyJAMBU learns moderately in the early rounds. In Round 1, logistic regression and random forest have an accuracy of about 71%. This implies that TinyJAMBU is learning perfect when the number of rounds is small. As the number of rounds increases, the accuracy gradually decreases. In Round 5, it is close to 50%, which means it is a random guess. Hamming distance increases gradually over the rounds. This defines that it becomes more difficult to make a prediction as the diffusion increases.

8.5 Analysis

Early rounds of TinyJAMBU are somewhat learnable. Accuracy decreases with increasing rounds. ML can learn a lightweight structure more easily. MLP slower but slightly better. Logistic regression effective and fast. Random forest stable performance

8.6 Conclusion

TinyJAMBU has a modest level of machine learning resistance. While fewer rounds can be learned, more rounds greatly increase security.

9. Trivium Cipher Analysis

9.1 About Trivium

Trivium is a lightweight stream cipher designed for hardware efficiency and low computational complexity. The cipher is a part of the eSTREAM project and is one of the final stream ciphers for good environments. The Trivium cipher makes use of three nonlinear feedback shift registers (NLFSR) and bitwise operations such as XOR, AND, and shift. However, unlike block ciphers such as AES and SIMON, the Trivium cipher generates

keystream, which is then XORed with the plaintext to produce the ciphertext. The Trivium cipher is well-suited for lightweight devices due to its register-based structure and bitwise operations. However, reduced round Trivium is prone to patterns and can be learned by machine learning.

9.2 Results

Table 7: Trivium – Bitwise Accuracy (%) across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	78.16	71.89	65.78	59.48	51.69
MLP	78.13	71.60	65.61	59.50	51.89
CNN	57.50	55.70	54.41	52.56	50.55
MINE	77.90	71.95	65.55	59.63	51.86
Random Forest	77.83	71.94	65.74	59.22	51.68

Table 8: Trivium – Hamming Distance across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	13.98	17.99	21.90	25.93	30.92
MLP	14.00	18.17	22.01	25.92	30.79
CNN	27.20	28.35	29.18	30.36	31.65
MINE	14.14	17.95	22.05	25.83	30.81
Random Forest	14.19	17.96	21.93	26.10	30.93

9.3 Plots

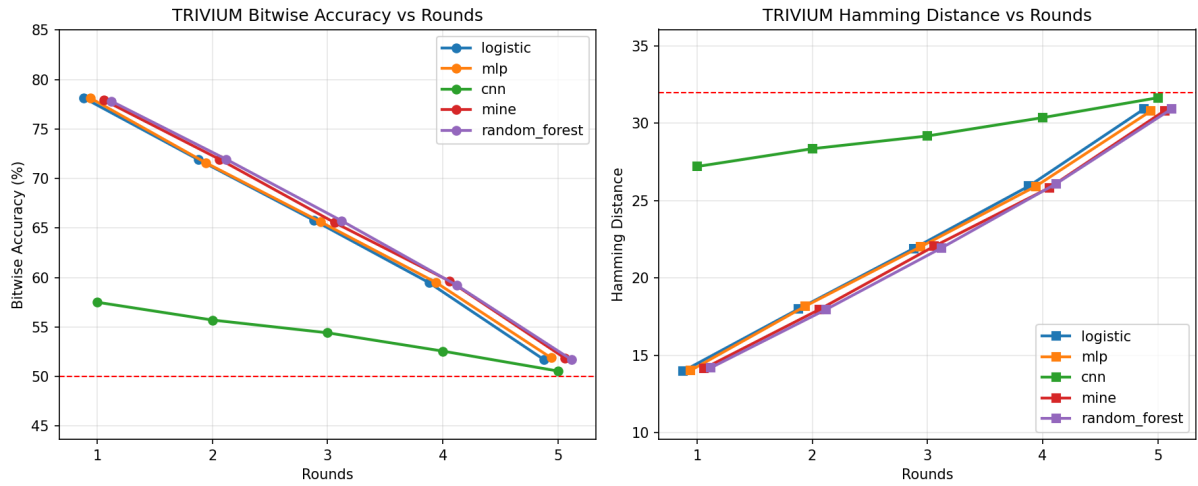


Figure 4: Trivium Accuracy vs Rounds and Hamming Distance vs Rounds

9.4 Observation

Accuracy of all models is near 78% in Round 1. Hence, the model learns better in state 1. As the number of rounds increases, the accuracy of the models drops. In the fifth round, the models are near random guessing at 50%. The hamming distances increase with the number of rounds. Hence, it is clear that the models are becoming increasingly difficult to predict as the complexity of the cipher also increases. The CNN model is near random guessing in all rounds. In the initial rounds, logistic regression and MLP are performing well.

9.5 Analysis

Trivium learnable in early rounds. With each round, accuracy declines. Lightweight structure easier to learn. Logistic regression performs well. MLP slower but slightly better. Random forest stable performance

9.6 Conclusion

Trivium's resistance to machine learning is modest. While fewer rounds can be learned, more rounds greatly increase the security of the cypher.

10. XOODOO Cipher Analysis

10.1 About XOODOO

XOODOO is a light mathematical process used in modern encryption that works by taking a block of data and mixing it up through various rounds of transformations to make it safe, yet fast enough to be performed on small machines. Due to its clean and structured approach, researchers are curious if machine learning approaches could identify hidden patterns in a weakened version of XOODOO that uses fewer rounds of scrambling compared to the usual rounds.

10.2 Results

Table 9: XOODOO – Bitwise Accuracy (%) across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	95.46	83.88	65.84	57.10	49.90
MLP	91.56	80.17	63.66	56.69	50.00
CNN	54.57	53.61	51.81	50.01	50.07
MINE	76.86	71.21	61.66	56.01	50.07
Random Forest	87.95	78.85	64.05	57.03	50.12

Table 10: XOODOO – Hamming Distance across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	2.91	10.32	21.86	27.45	32.07
MLP	5.40	12.69	23.26	27.72	32.00
CNN	29.07	29.69	30.84	31.99	31.96
MINE	14.81	18.43	24.54	28.16	31.95
Random Forest	7.71	13.54	23.01	27.50	31.93

10.3 Plots

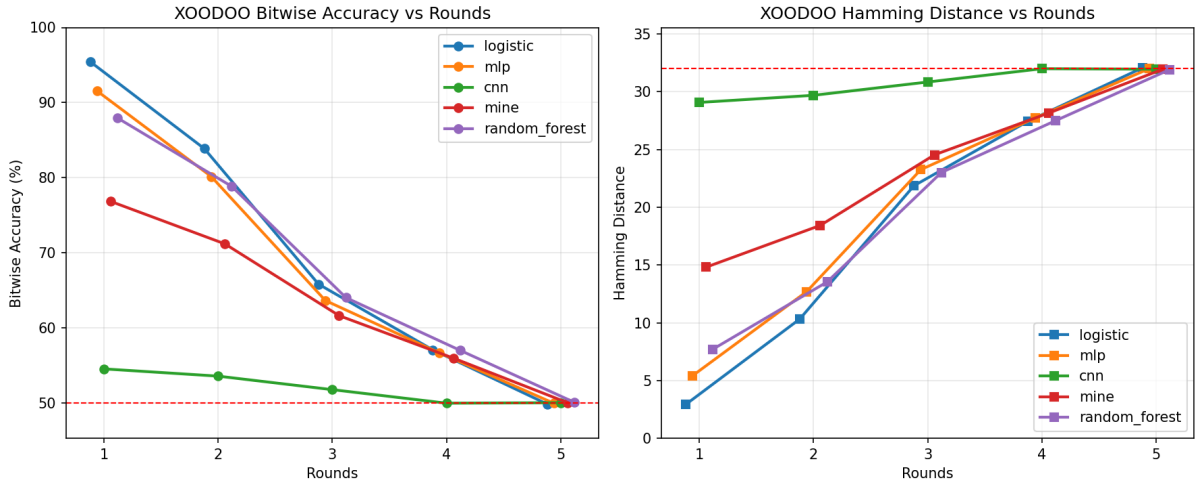


Figure 5: XOODOO Accuracy vs Rounds and Hamming Distance vs Rounds

10.4 Observation

XOODOO learns good in early rounds. In Round 1, logistic regression achieves 95.46% accuracy, which is the highest among all evaluated ciphers so far. This indicates that reduced-round XOODOO contains strong learnable patterns. As rounds increase, accuracy decreases rapidly. By Round 5, accuracy approaches 50%, indicating random guessing. Hamming distance increases steadily with rounds, confirming that predictions become harder as diffusion increases. CNN performs near random guessing across all rounds, while logistic regression performs best in early rounds.

10.5 Analysis

XOODOO highly learnable in early rounds. Highest accuracy observed among ciphers. Rapid accuracy drop with rounds. Logistic regression give best result. MLP works efficiently but more slowly. Random forest performance is stable.

10.6 Conclusion

With fewer rounds, machine learning models can easily pick up patterns in XOODOO. But as the number of rounds increases, the data gets more thoroughly scrambled and

mixed, making it nearly impossible for any AI model to detect any useful patterns or crack the encryption.

11. PRESENT Cipher Analysis

11.1 About PRESENT

PRESENT is a lightweight block cipher designed specifically for constrained environments such as RFID tags, IoT devices, and embedded systems. PRESENT works on a 64-bit block size and supports 80-bit or 128-bit keys. PRESENT is based on a substitution-permutation network similar to AES but with a significantly simpler structure. It uses small S-boxes and permutation layers to achieve diffusion and confusion. Due to its lightweight design, reduced-round PRESENT may expose patterns that machine learning models can learn. PRESENT typically uses 31 rounds in standard implementations, but in this project we evaluate reduced rounds to analyze learnability.

11.2 Results

Table 11: PRESENT – Bitwise Accuracy (%) across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	73.93	61.18	57.03	54.06	50.76
MLP	96.40	61.54	56.74	54.07	50.62
CNN	54.68	52.45	51.69	51.36	50.40
MINE	65.55	59.10	56.50	53.79	50.64
Random Forest	69.96	60.17	56.50	53.95	50.62

Table 12: PRESENT – Hamming Distance across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	16.69	24.84	27.50	29.40	31.51
MLP	2.30	24.62	27.69	29.40	31.60
CNN	29.01	30.43	30.92	31.13	31.75
MINE	22.05	26.18	27.84	29.57	31.59
Random Forest	19.22	25.49	27.84	29.47	31.60

11.3 Plots

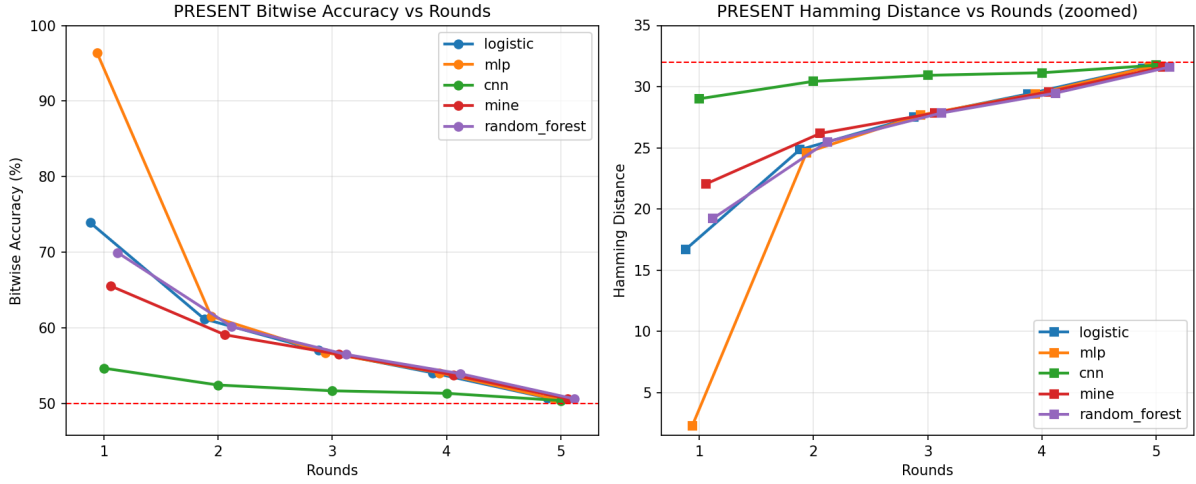


Figure 6: PRESENT Accuracy vs Rounds and Hamming Distance vs Rounds

11.4 Observation

PRESENT learns well in Round 1. MLP achieves 96.40% accuracy, which is one of the highest among all evaluated ciphers. This indicates that reduced-round PRESENT contains strong learnable patterns. However, accuracy drops significantly from Round 2 onwards. By Round 5, accuracy approaches 50%, indicating random guessing. Hamming distance increases across rounds, showing that predictions become less accurate as cipher complexity increases. CNN performs close to random guessing across all rounds. MLP performs best in early rounds but requires significantly higher training time.

11.5 Analysis

PRESENT highly learnable in Round 1. Sharp accuracy drop after Round 2. Lightweight structure easier to learn. MLP highest accuracy but slowest. Logistic regression efficient and fast. Random forest stable performance

11.6 Conclusion

PRESENT cipher shows strong learnability in reduced rounds. However, increasing rounds improves diffusion and security, making the cipher resistant to machine learning based attacks.

12. KATAN Cipher Analysis

12.1 About KATAN

KATAN is an lightweight block cipher designed for hardware environments such as RFID tags, sensor networks, and embedded devices. KATAN was specifically designed to minimize hardware cost and power consumption. KATAN uses shift registers and

simple nonlinear functions to perform encryption. The cipher operates using a Feistel-like structure and relies on bitwise operations such as XOR and AND. Due to its lightweight design and simple structure, reduced-round KATAN may expose learnable patterns.

12.2 Results

Table 13: KATAN – Bitwise Accuracy (%) across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	72.04	64.40	57.57	52.93	50.18
MLP	72.03	64.03	57.85	53.08	50.12
CNN	60.84	57.26	53.72	51.58	49.73
MINE	71.75	64.26	57.52	53.24	49.81
Random Forest	71.95	64.01	57.77	53.14	49.88

Table 14: KATAN – Hamming Distance across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	8.95	11.39	13.58	15.06	15.94
MLP	8.95	11.51	13.49	15.02	15.96
CNN	12.53	13.68	14.81	15.50	16.09
MINE	9.04	11.44	13.59	14.96	16.06
Random Forest	8.98	11.52	13.51	15.00	16.04

12.3 Plots

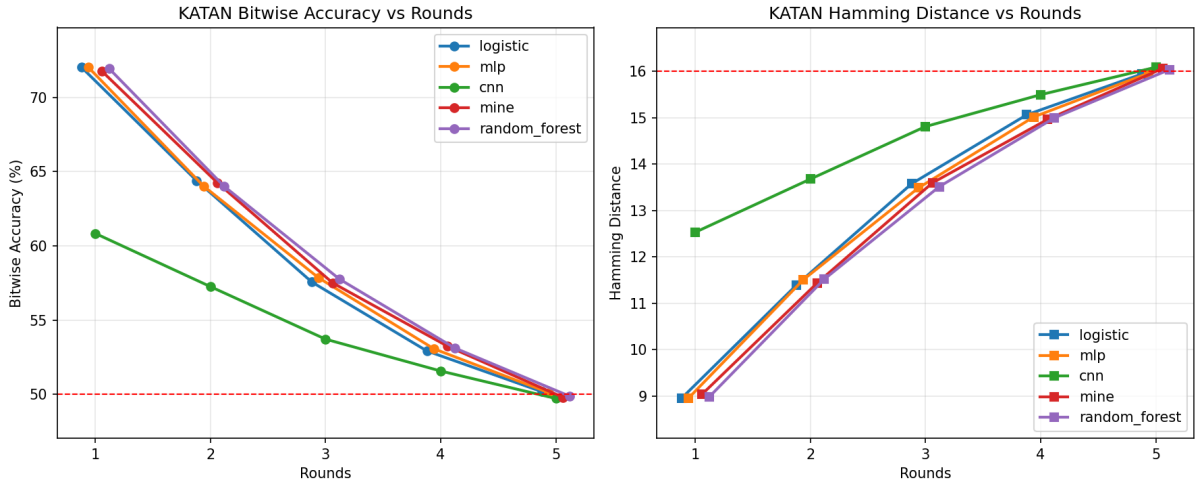


Figure 7: KATAN Accuracy vs Rounds and Hamming Distance vs Rounds

12.4 Observation

KATAN shows moderate learnability in early rounds. In Round 1, all models achieve around 72% accuracy, indicating that reduced-round KATAN contains learnable patterns.

Accuracy decreases gradually as rounds increase. By Round 5, accuracy approaches 50%, indicating random guessing. Hamming distance increases steadily across rounds, showing that prediction becomes harder as diffusion increases. CNN performs slightly worse compared to other models. Logistic regression and random forest perform consistently well.

12.5 Analysis

KATAN moderately learnable in early rounds. Gradual accuracy decrease with rounds. Lightweight design introduces learnable patterns. Logistic regression fast and effective. Random forest stable performance. MLP slightly slower but comparable

12.6 Conclusion

KATAN cipher shows moderate resistance to machine learning. Reduced rounds are learnable, but increasing rounds improves cipher security.

13. SPECK Cipher Analysis

13.1 About SPECK

SPECK is a lightweight block cipher designed by the National Security Agency (NSA) for efficient software implementation. Unlike SIMON, which is optimized for hardware, SPECK is designed for software environments such as microcontrollers and embedded processors.

SPECK is based on ARX operations, which include:

- Addition
- Rotation
- XOR

These operations are computationally efficient and widely supported across platforms. SPECK uses a Feistel-like structure with multiple rounds to achieve diffusion and confusion.

13.2 Results

Table 15: SPECK – Bitwise Accuracy (%) across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	71.99	67.24	62.35	57.53	52.85
MLP	72.86	67.19	62.40	57.36	49.91
CNN	65.22	62.52	59.09	55.79	52.96
MINE	71.90	67.08	62.53	57.54	52.84
Random Forest	79.91	67.36	62.68	57.78	53.05

Table 16: SPECK – Hamming Distance across Rounds 1–5

Model	Round 1	Round 2	Round 3	Round 4	Round 5
Logistic Regression	8.96	10.48	12.05	13.59	15.09
MLP	8.69	10.50	12.03	13.65	16.03
CNN	11.13	11.99	13.09	14.15	15.05
MINE	8.99	10.53	11.99	13.59	15.09
Random Forest	6.43	10.45	11.94	13.51	15.02

13.3 Plots

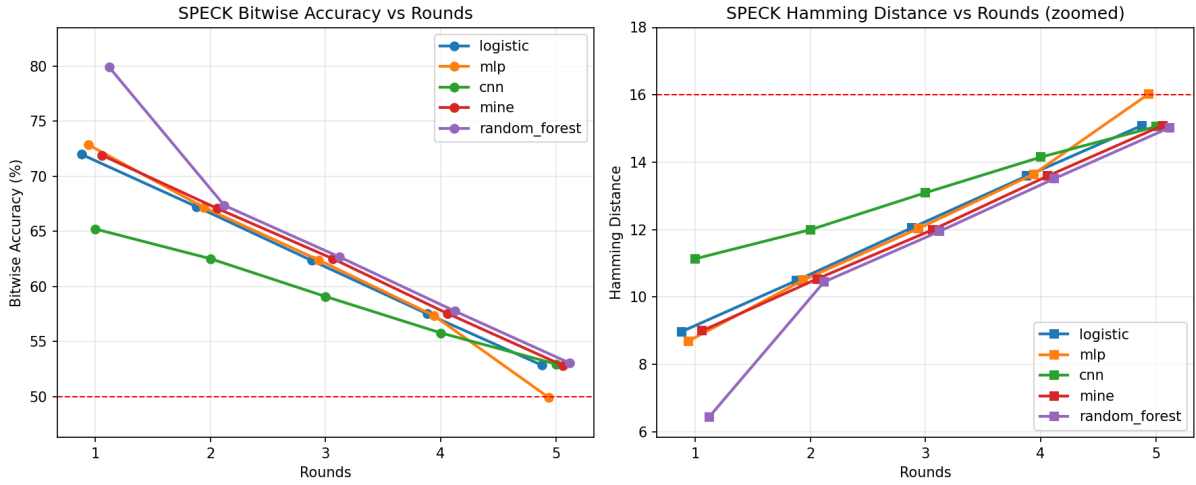


Figure 8: SPECK Accuracy vs Rounds and Hamming Distance vs Rounds

13.4 Observation

SPECK shows moderate learnability in early rounds. In Round 1, random forest achieves the highest accuracy of 79.91%, which tells that reduced-round SPECK contains learnable patterns. Accuracy decreases as rounds increase. By Round 5, accuracy approaches 50%, indicating random guessing. Hamming distance increases steadily with rounds, confirming increasing diffusion and improved cipher security.

13.5 Analysis

SPECK moderately learnable in early rounds. Gradual accuracy decrease with rounds. ARX structure introduces learnable patterns. Random forest performs best in early rounds. Logistic regression fast and stable. MLP slower with comparable performance

13.6 Conclusion

SPECK shows moderate resistance to machine learning. Reduced rounds are learnable, but increasing rounds significantly improves cipher security.

14. Final Analysis

This project compares eight cryptographic ciphers using machine learning based algorithm. The ciphers analyzed include AES, SIMON, TinyJAMBU, Trivium, XOODOO, PRESENT, KATAN, and SPECK. The performance of different machine learning models was analyzed using bitwise accuracy and Hamming distance across multiple reduced rounds.

14.1 Accuracy Trend Across Ciphers

A consistent trend was observed across all eight ciphers:

- Bitwise accuracy is highest in early rounds and decreases with increasing round count.
- All ciphers approach approximately 50% accuracy by Round 5.
- Hamming Distance increases steadily across rounds for all ciphers.

This universal behavior confirms that reduced-round ciphers retain learnable statistical structure, while increasing rounds introduces stronger diffusion and confusion that progressively eliminates this structure.

14.2 Cipher Learnability Ranking

Ranked by best Round 1 accuracy, from most to least learnable:

1. PRESENT – 96.40% (MLP): highest learnability, driven by 4-bit S-box insufficiency at Round 1
2. XOODOO – 95.46% (Logistic): very high linear learnability from a single permutation round
3. SIMON – 93.65% (MLP): very high learnability from slow Feistel diffusion accumulation
4. SPECK – 79.91% (RF): moderate-high learnability; ensemble methods outperform others
5. Trivium – 78.16% (Logistic): high but purely linear leakage from partial NLFSR initialization
6. TinyJAMBU – 71.89% (RF): moderate, uniformly distributed across model types
7. KATAN – 72.04% (LR): moderate, near-perfect uniformity across all models
8. AES – 70.30% (MLP): lowest learnability; strongest resistance among all evaluated ciphers

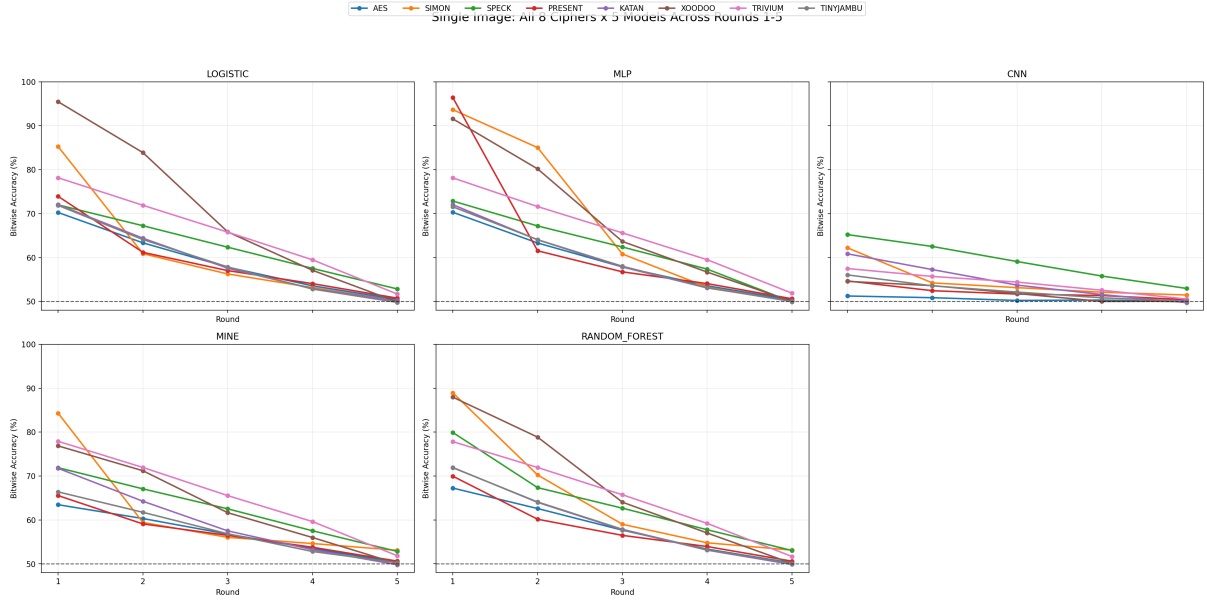


Figure 9: All cipher and model comparison

14.3 Model Performance Comparison

- **MLP** achieved the highest overall accuracy for most ciphers due to its nonlinear expressive capacity, at the cost of significantly longer training times.
- **Logistic Regression** matched or exceeded MLP on linearly-structured ciphers (Trivium, XOODOO, KATAN, TinyJAMBU, AES), making it the most computationally efficient model overall.
- **Random Forest** was the best-performing model on SPECK and delivered consistently stable results across all ciphers.
- **MINE** consistently occupied the middle performance tier, providing complementary information-theoretic leakage estimates.
- **CNN** underperformed across virtually all ciphers and rounds, typically 5–15 percentage points below the best model. Local feature extraction does not align with the global non-spatial structure of ciphertext bit patterns.

14.4 Hamming Distance Analysis

Hamming distance increased with rounds for all ciphers. This indicates:

- Increased diffusion
- Reduced prediction accuracy
- Stronger cipher security

Lower Hamming distance in early rounds confirmed learnable patterns in reduced-round ciphers.

14.5 Overall Observations

- Lightweight ciphers easier to learn
- Reduced rounds vulnerable to ML attacks
- Increasing rounds improves security
- AES strongest among evaluated ciphers
- XOODOO most learnable in early rounds
- CNN least effective model

15. Conclusion

In this project, machine learning based analysis was performed on eight cryptographic ciphers. Multiple machine learning models including Logistic Regression, Multi Layer Perceptron, CNN, MINE, and Random Forest were used to evaluate cipher learnability. The results indicates that reduced-round lightweight ciphers exhibit learnable patterns that machine learning models can exploit. However, as the number of rounds increases, cipher diffusion improves and model accuracy approaches random guessing.

Among machine learning models, MLP achieved the highest accuracy but required longer training time. Logistic regression and random forest provided stable and efficient performance.

This study tells that machine learning based cryptanalysis can be used to evaluate cipher security and understand the impact of reduced rounds. Increasing cipher rounds improves resistance against machine learning attacks.

15.1 Future Work

Future work may include:

- Extension to full-round cipher configurations with larger datasets.
- Evaluation of cipher implementations under hardware noise to extend the ML threat model to physical attack scenarios.
- Exploring deep learning architectures.
- Testing additional lightweight ciphers.
- Evaluating real-world cryptographic implementations.

16. References

1. A. Alani, "Machine Learning Based Cryptanalysis of Reduced Round Block Ciphers", IEEE, 2015. Available: <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=7148500>

2. R. Beaulieu et al., "The SIMON and SPECK Lightweight Block Ciphers", IACR Cryptology ePrint Archive, 2012. Available: <https://eprint.iacr.org/2012/529.pdf>
3. A. Sharma et al., "Machine Learning Based Cryptanalysis: A Survey", Mathematics Journal, MDPI, 2021. Available: <https://www.mdpi.com/2227-7390/9/22/2970>
4. National Institute of Standards and Technology (NIST), "Advanced Encryption Standard (AES)", FIPS Publication 197, 2001.
5. D. R. Stinson, "Cryptography: Theory and Practice", SAGE Journals. Available: <https://journals.sagepub.com/doi/pdf/10.1177/0270467603251296>
6. TinyJAMBU Team, "TinyJAMBU Lightweight Authenticated Encryption", IACR Cryptology ePrint Archive, 2018. Available: <https://eprint.iacr.org/2018/767.pdf>
7. L. R. Knudsen, "Truncated and Higher Order Differentials", ACM Conference Proceedings. Available: <https://dl.acm.org/doi/abs/10.1145/169701.169696>
8. X. Lai, J. Massey, "Markov Ciphers and Differential Cryptanalysis", World Scientific Journal. Available: <https://www.worldscientific.com/doi/abs/10.1142/S0218127497001618>
9. Goodfellow, Ian, Bengio, Yoshua, Courville, Aaron, "Deep Learning", MIT Press, 2016.
10. R. Beaulieu et al., "The SIMON and SPECK Lightweight Block Ciphers", Proceedings of the 52nd Annual Design Automation Conference, 2015.
11. TinyJAMBU Team, "TinyJAMBU Lightweight Authenticated Encryption", NIST Lightweight Cryptography Project.